# Huawei HCCDA-AI certification

Instructor : Fawad Bahadur

# Machine Learning Basics

## Machine Learning Algorithms and Applications in Python

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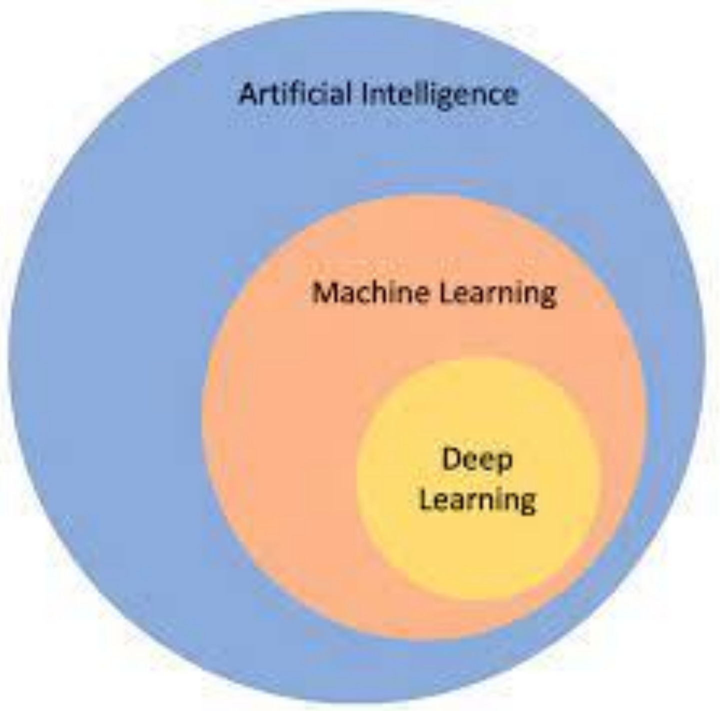
# What is Machine Learning?

Machine Learning (ML) is the part of AI that lets computers learn from data and get better on their own—no step-by-step programming needed.

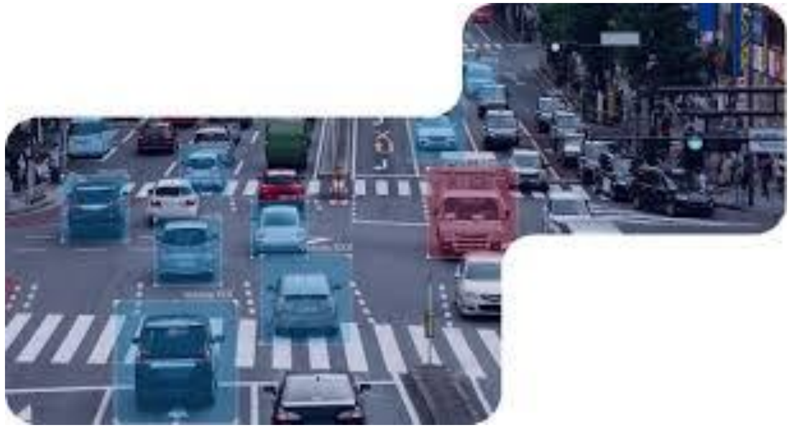



"Machine learning is the science of getting computers to act without being explicitly programmed." – **Arthur Samuel**

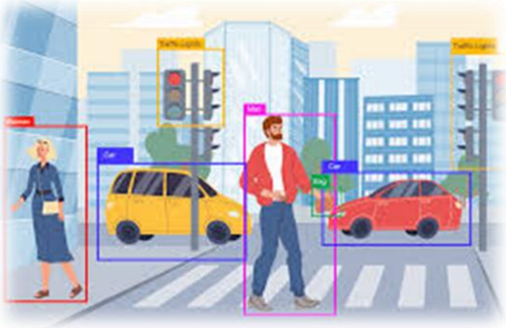
Data → Learning Algorithm → Model → Predictions

# Where ML is Used Today

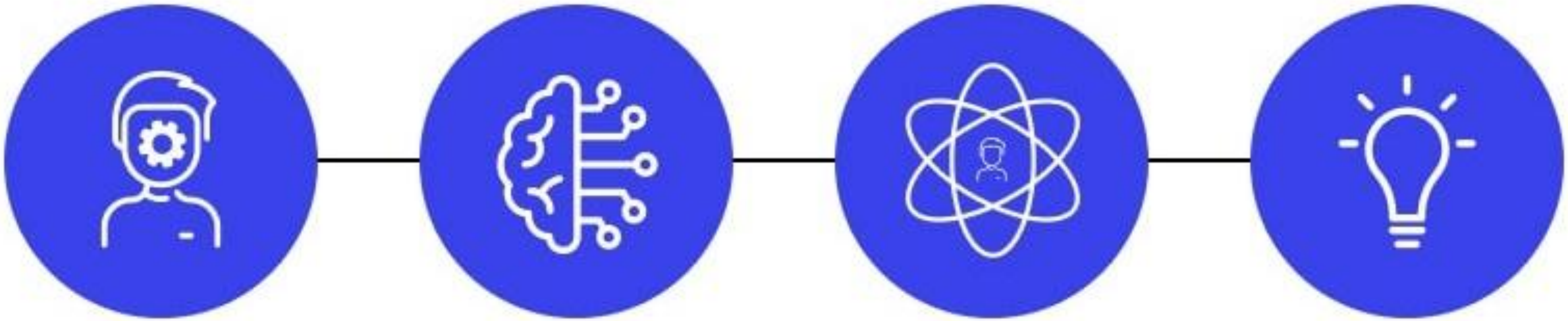
1. Face recognition
2. Object detection
3. Chatbots
4. Recommendation systems
5. Autonomous vehicles
6. Disease diagnosis
7. Sentiment analysis
8. Yield estimation
9. Fraud detection





# Types of Machine Learning



Supervised machine learning

Unsupervised machine learning

Semi-supervised learning

Reinforcement learning

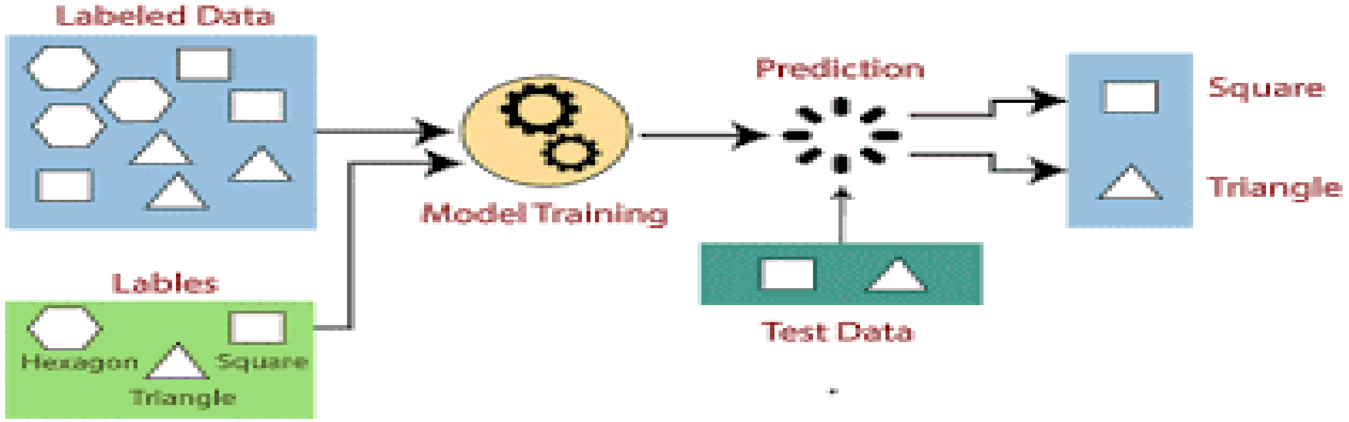
# Supervised Learning

**Definition:**
The model learns from labeled data to make predictions.

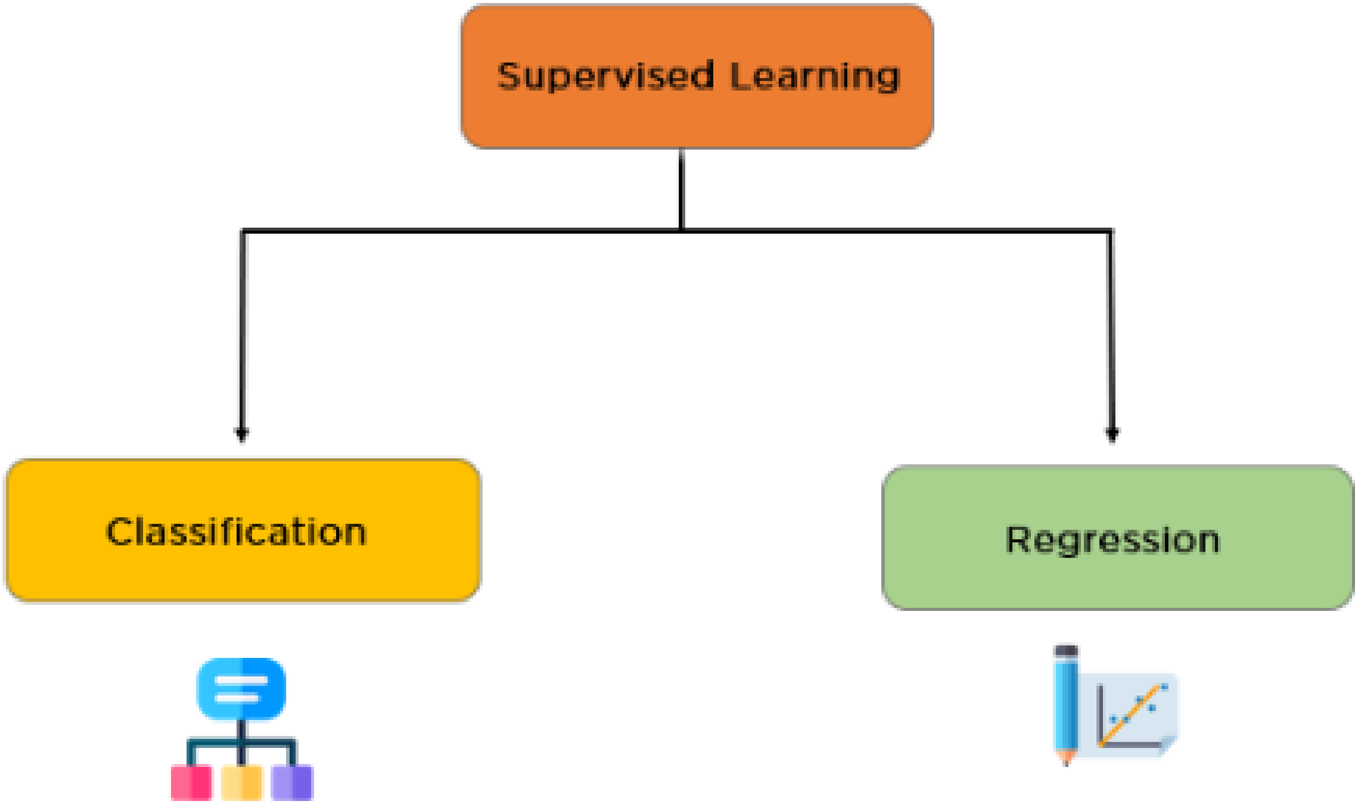
X → Y

**Examples:**
Spam email detection, house price prediction, image classification.

| Size (x) | Price (y) |
|---|---|
| 2104 | 460 |
| 1416 | 232 |
| 1534 | 315 |
| 852 | 178 |
| … | … |



# Types of Supervised Learning Tasks



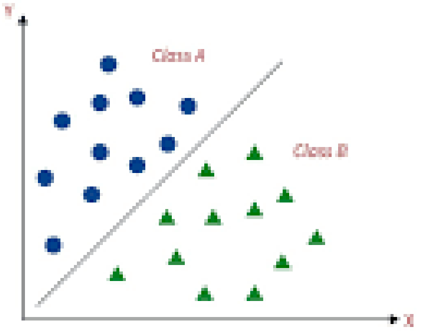
# Classification

**Definition:**
Predicts discrete labels or categories.

**Goal:**
Assign an input to one of several predefined classes.

**Examples:**

- Email spam detection (Spam / Not Spam)
- Image recognition (Pedestrian, Car, Motorcycle or Truck)



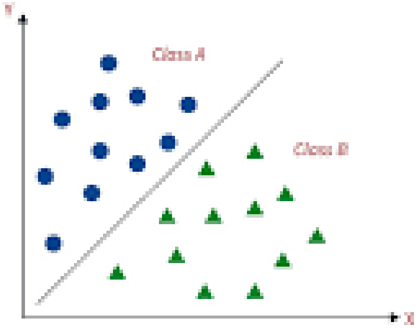


Pedestrian Car Motorcycle Truck

# Binary Classification

**Definition:**

Classification task with two possible outcomes.




**Label Format:**

Usually 0 and 1 (or True/False)

**Examples:**

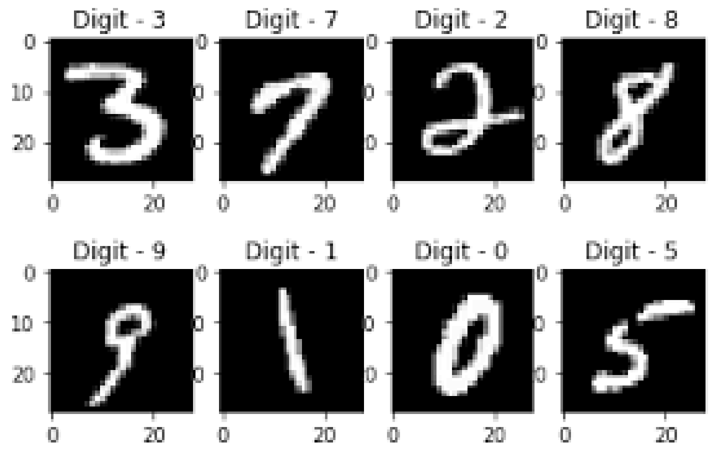
- Spam vs. Not Spam
- Disease: Positive vs. Negative
- Pass vs. Fail
- Real vs Fake

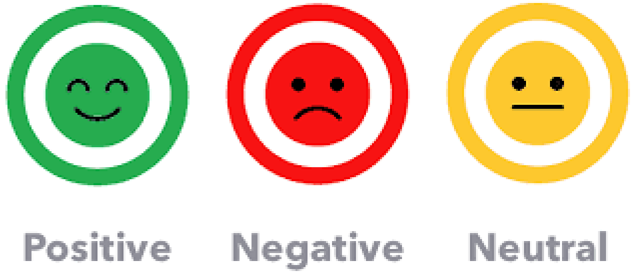
# Multi-Class Classification

**Definition:**
Classification task with more than two classes.

**Label Format:**
Classes like 0, 1, 2, ..., or Cat, Dog, Horse

**Examples:**
- Handwritten digit recognition (0–9)
- Animal type (Cat, Dog, Horse)
- Sentiment analysis (Negative, Neutral, Positive)



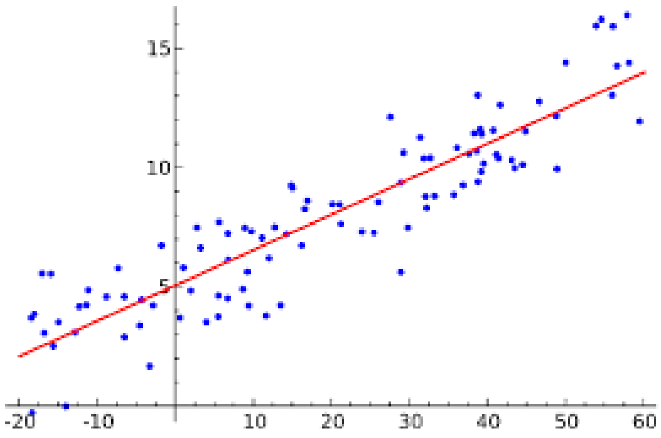



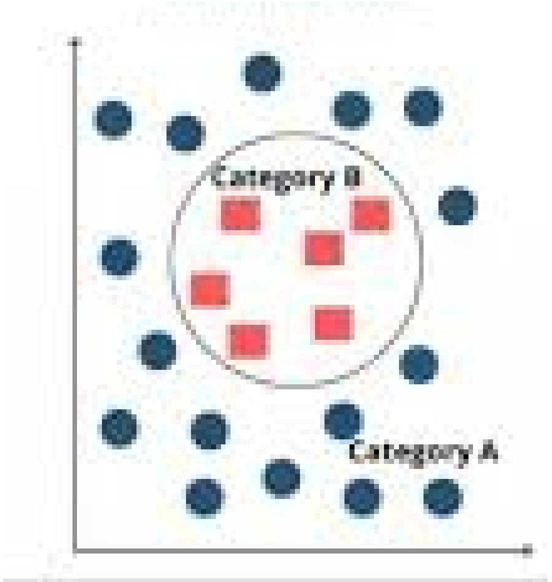
# Regression

**Definition:**
Predicts continuous numerical values.

**Goal:**
Estimate a real value based on input features.

**Examples:**
- Predicting house prices
- Forecasting temperature



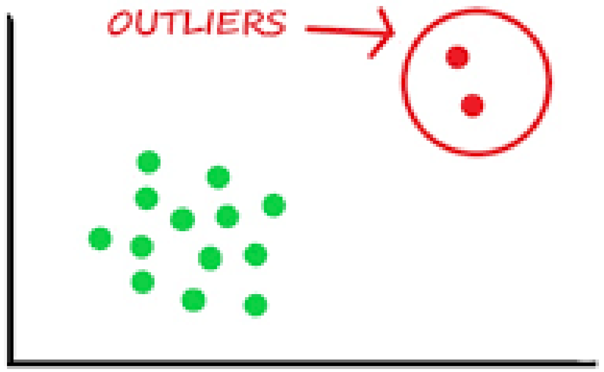
# Understanding Outliers

An outlier is a data point that significantly **differs** from other observations in the dataset.

It **doesn't** follow the **pattern** of the rest of the data.

**Examples**

- A person earning $10 million in a dataset where others earn $30k–$80k.
- Temperature reading of 1000°C in a normal weather dataset.

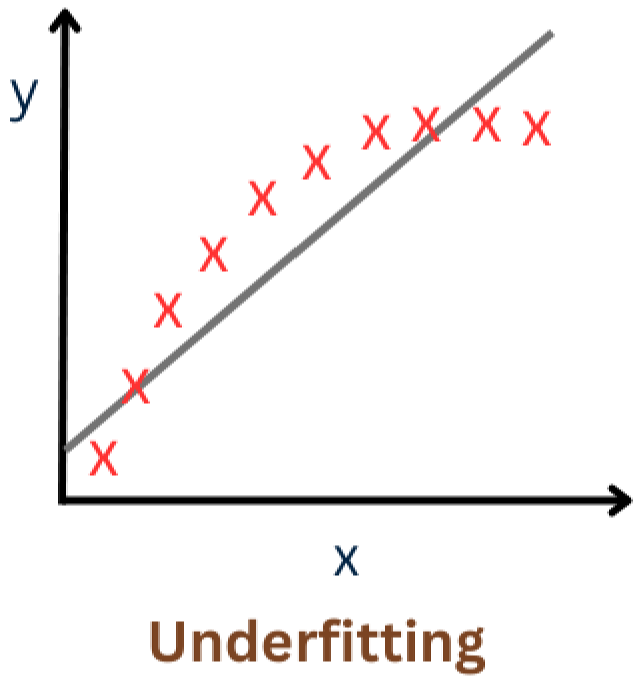


# Underfitting

Model is too simple → misses even obvious patterns.

**Symptoms:**

- Low accuracy on both training & test sets (unseen data)
- Flat learning curves (no improvement with more data)




**Example:**

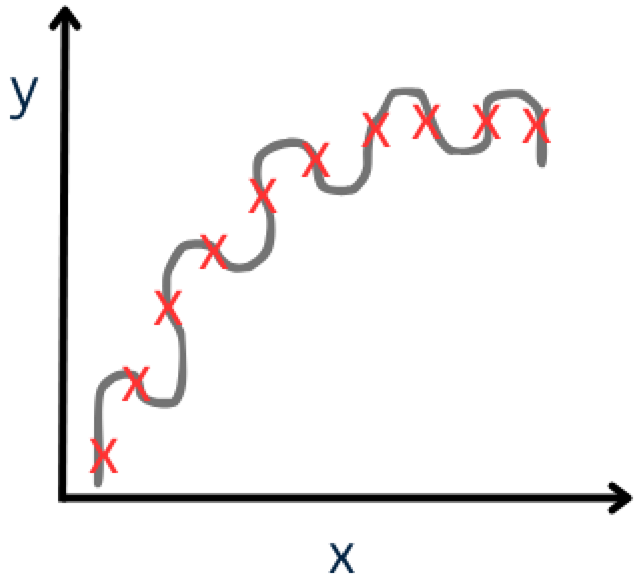
Predicting house price based on area, number of rooms, location score, etc.

# Overfitting

Model is too complex → memorizes noise & outliers.

**Symptoms:**

- High training accuracy, poor test accuracy
- Validation loss rises while training loss keeps dropping



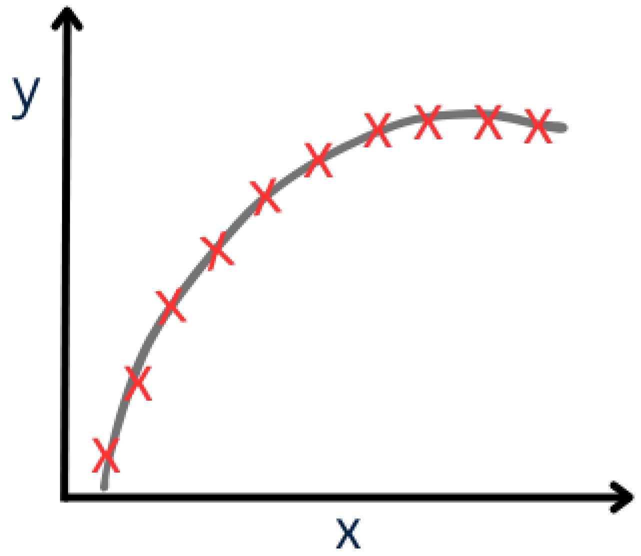

Overfitting

# Just-Right Fit

Captures true pattern, ignores noise.

**Symptoms:**

- Similar, high performance on train & test sets




- Overfitting is memorizing;
- underfitting is ignoring.
- Good modeling is learning.

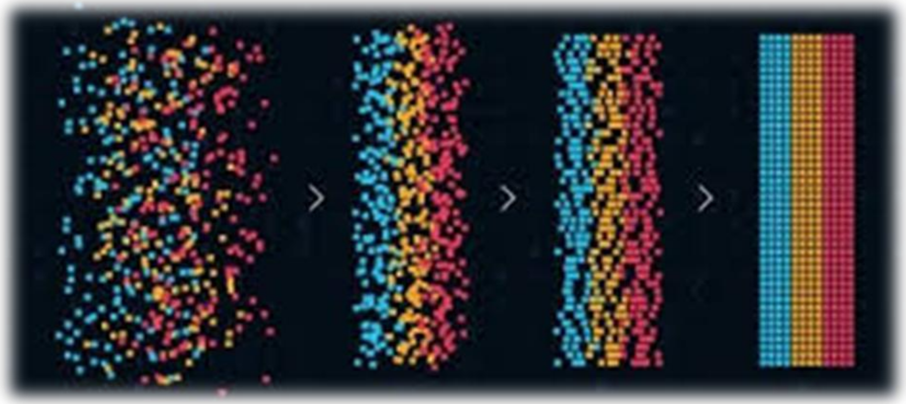
# Unsupervised Learning

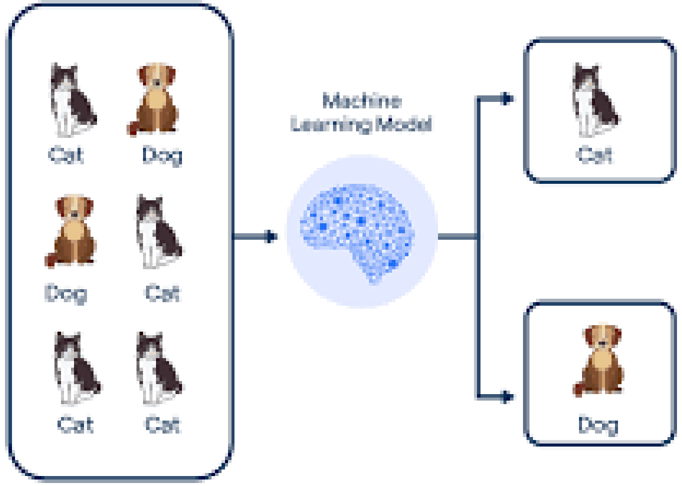
**Definition:**

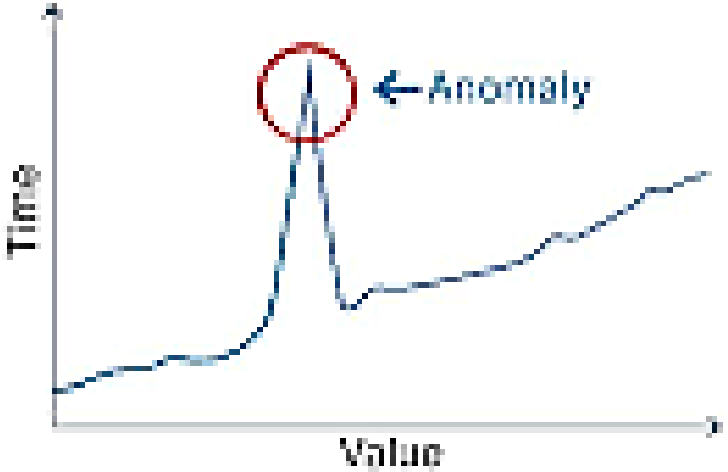
The model finds patterns or groupings in unlabeled data.

**Examples:**

Anomaly detection, Pattern Recognition





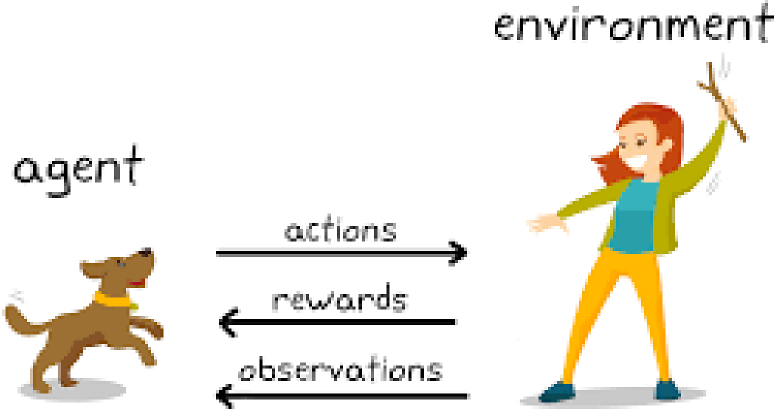
# Reinforcement Learning

**Definition:**

The model learns by interacting with an environment and receiving rewards or penalties.

**Examples:**

Game-playing AI (e.g., AlphaGo), robotic navigation.




# Types of Machine Learning

| Type | Core Idea | Typical Data | Use-Cases | Key Algorithms |
|---|---|---|---|---|
| **Supervised** | Learn from labeled examples (input + correct output). | Labeled data: e.g., images tagged "cat" or "dog". | Spam detection | Linear / Logistic regression, Random forest, SVM |
| **Unsupervised** | Discover hidden patterns without labels. | Unlabeled data. | Customer segmentation | K-Means, DBSCAN, Apriori |
| **Semi-Supervised** | Small labeled set + large unlabeled set. | Mix of the above. | Fraud detection | Self-training, Label propagation |
| **Reinforcement** | Agent learns by trial-and-error via rewards. | Environment + reward signals. | Game playing | Q-Learning, Deep RL |

# THANK YOU